Race to Idle: Why Efficiency Cores Do Not Save Energy for LLM Decode on x86 Heterogeneous SoCs

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Artifact
The measurement harness, the complete dataset including every discarded run, and the analysis that generates every table and figure in the paper are archived at doi:10.5281/zenodo.23243517. This link is withheld from the main manuscript because the repository identifies the author. It will appear in the camera-ready version.

Biography
Muhammad Etisam Ul Haq received the B.S. degree in data science from the Punjab University College of Information Technology (PUCIT), University of the Punjab, Lahore, Pakistan, and is currently pursuing the M.S. degree in data science at the Institute of Data Science, University of Engineering and Technology, Lahore, Pakistan.
He works as a machine learning engineer, where his practice covers large language model fine-tuning and inference serving, retrieval-augmented generation, and private inference on non-CUDA accelerators. His research interests include the energy efficiency of on-device machine learning, measurement methodology for heterogeneous processors, and the systems side of language model deployment.

Declarations
Conflicts of interest: none.
Funding: this work received no external funding.
AI disclosure: as stated in the manuscript's Acknowledgment section, Anthropic's Claude (Claude Opus 5) was used as a writing and engineering assistant for manuscript text, analysis code and figure generation. The research questions, experimental design, measurements and interpretation are the author's own.